CLASS 8 · CBSE · GANITA PRAKASH PART 2 · SHORT NOTES

Chapter 6: Algebra Play

Topic-wise short notes with solved examples

Using algebra to explain number tricks and puzzles: "think of a number" tricks, number pyramids, calendar grids, largest products, divisibility tricks with digits, and word problems solved with equations.

1 Think of a number

- In a "think of a number" trick, call the unknown number x and write each step as an expression. Simplifying shows why the trick works.
- If x **cancels out**, everyone gets the same answer. If x remains, the answer reveals the starting number.

Example 1.1 Think of a number. Double it. Add 4. Divide by 2. Subtract the number you first thought of. Why is the answer always 2?

SOLUTION

$x \to 2x \to 2x + 4 \to x + 2 \to x + 2 - x$

The x cancels and 2 is left, whatever the starting number

Example 1.2 Change the trick above so that the answer is always 5.

SOLUTION

The final answer is half of the number added in the third step.

Add 10 instead of 4

Example 1.3 Think of a number. Multiply by 3. Add 12. Divide by 3. Subtract 4. What do you get?

SOLUTION

$x \to 3x \to 3x + 12 \to x + 4 \to x$

The number you started with

2 The date trick

- A trick can hide two numbers in one answer by using place value.
- Steps: take the month M; multiply by 4; add 7; multiply by 25; add the day D.
- Algebra: $M \to 4M \to 4M + 7 \to 100M + 175 \to 100M + 175 + D$.
- Subtract 175 from the answer. Because D is at most 31, the last two digits are the day and the digits before them are the month.

Example 2.1 A friend follows the steps above and reports 990. Find the date.

SOLUTION

$990 - 175 = 815$

Month 8, day 15: 15 August

Example 2.2 The reported answer is 1400. Find the date.

SOLUTION

$1400 - 175 = 1225$

25 December

- Any steps that lead to $100M + k + D$ will work. Only the number k to be subtracted changes.

3 Number pyramids

- In a number pyramid each number is the **sum of the two numbers directly below it**.
- Missing numbers lower down can be found by subtraction, or by using letters and solving an equation.
- **Top number in terms of the bottom row**:

Rows	Bottom row	Top number
2	a, b	$a + b$
3	a, b, c	$a + 2b + c$
4	a, b, c, d	$a + 3b + 3c + d$
5	a, b, c, d, e	$a + 4b + 6c + 4d + e$

- The multipliers 1; 1, 1; 1, 2, 1; 1, 3, 3, 1; 1, 4, 6, 4, 1 form the rows of a well-known number triangle. Numbers in the middle of the bottom row count for more than those at the ends.

Example 3.1 The bottom row of a 3-row pyramid is 8, 13, 4. Find the top number without building the pyramid.

SOLUTION

$8 + 2 \times 13 + 4$

38

Example 3.2 The bottom row of a 4-row pyramid is 7, 18, 6, 5. Find the top number.

SOLUTION

$7 + 3 \times 18 + 3 \times 6 + 5$

84

Example 3.3 A 3-row pyramid has 60 at the top and 12, c, 8 in the bottom row. Find c.

SOLUTION

$12 + 2c + 8 = 60$, so $2c = 40$

$c = 20$ **(the middle row is 32 and 28)**

Example 3.4 The bottom row of a 3-row pyramid holds the Virahāṅka–Fibonacci numbers 1, 2, 3. Fill in the pyramid.

SOLUTION

Middle row: 3 and 5. Top: 8.

All the numbers in the pyramid are Virahāṅka–Fibonacci numbers

4 Calendar and number grids

- In a calendar, the number to the right is 1 more and the number below is 7 more.
- A 2×2 block with top-left number a holds a, $a + 1$, $a + 7$, $a + 8$. Its sum is $4a + 16$.
- So from the sum alone the whole block can be found: $a = \dfrac{\text{sum} - 16}{4}$.
- A 3×3 block with centre m has sum $9m$.
- On a 1-to-100 number grid with 10 in a row, the number below is 10 more, so a 2×2 block has sum $4a + 22$.

Example 4.1 The four dates in a 2×2 block of a calendar add up to 76. Find them.

SOLUTION

$4a + 16 = 76$, so $a = 15$

15, 16, 22, 23

Example 4.2 Nine dates in a 3×3 block add up to 144. Find the middle date and the smallest date.

SOLUTION

$9m = 144$, so $m = 16$. The smallest is $m - 8$.

16 and 8

Example 4.3 Can the four numbers of a 2×2 calendar block add up to 50?

SOLUTION

$4a + 16 = 50$ gives $a = 8.5$

No. The sum must be 16 more than a multiple of 4.

5 Algebra grids

- When shapes or letters stand for numbers and row or column sums are given, write an equation for each row and combine them.
- Start with a row that has only one kind of shape.

Example 5.1 In a grid, three circles add up to 27, and two circles and a square add up to 19. Find the value of each shape.

SOLUTION

$3c = 27$ gives $c = 9$. Then $18 + s = 19$.

Circle 9, square 1

Example 5.2 Two triangles and a star add up to 21. One triangle and two stars add up to 15. Find each.

SOLUTION

Adding the equations: $3t + 3s = 36$, so $t + s = 12$. Subtracting this from the first: $t = 9$.

Triangle 9, star 3

6 The largest product

- Three different digits $p < q < r$ are placed in the pattern (two-digit number) $\times$ (one-digit number). There are 6 arrangements.
- For a fixed multiplier, the larger two-digit number gives the larger product. This leaves $rq \times p$, $rp \times q$ and $qp \times r$.
- Comparing $qp \times r = 10qr + pr$ with $rp \times q = 10qr + pq$: the first is larger because $r > q$.
- **Rule**: use the **largest digit as the multiplier** and arrange the other two in decreasing order.

Example 6.1 Use 2, 3 and 5 once each in the pattern two-digit $\times$ one-digit to get the largest product.

SOLUTION

Largest digit 5 is the multiplier; the others form 32.

$32 \times 5 = 160$

Example 6.2 Do the same with 1, 3 and 7.

SOLUTION

$31 \times 7 = 217$ **(compare** $71 \times 3 = 213$**)**

7 Divisibility tricks with digits

- A two-digit number with digits a and b is $10a + b$. A three-digit number is $100a + 10b + c$.

Trick	Algebra	Result
Two-digit number minus its reverse	$(10a+b)-(10b+a)=9(a-b)$	Divisible by 9; quotient is the difference of the digits
Two-digit number plus its reverse	$(10a+b)+(10b+a)=11(a+b)$	Divisible by 11; quotient is the sum of the digits
Three-digit number minus its reverse	$99(a-c)$	Divisible by 99
$abc+bca+cab$	$111(a+b+c)=37\times 3\times(a+b+c)$	Divisible by 37 and by 3
Six-digit number $abcabc$	$abc\times 1001=abc\times 7\times 11\times 13$	Dividing by 7, 11 and 13 gives back abc

Example 7.1 Without dividing, find the quotient when $83-38$ is divided by 9 and when $83+38$ is divided by 11.

SOLUTION

Difference of digits $=5$; sum of digits $=11$.

5 and 11

Example 7.2 A three-digit number is written twice to make 528528. What do you get on dividing by 7, then 11, then 13?

SOLUTION

$528528=528\times 1001$ and $1001=7\times 11\times 13$

528

8 Word problems with equations

- Choose a letter for the unknown, translate each sentence into algebra, form an equation and solve. Check the answer in the original words.
- Some problems can also be solved by reasoning without letters, for example by **supposing** all the animals are of one kind.

Example 8.1 A farm has goats and hens. There are 40 heads and 110 legs. How many of each are there?

SOLUTION

If all 40 were hens there would be 80 legs. The extra 30 legs come from goats, each adding 2 legs.

15 goats and 25 hens

Example 8.2 A mother is 4 times as old as her daughter. In 5 years she will be 3 times as old. Find the daughter's age.

SOLUTION

$4d + 5 = 3(d + 5)$ gives $4d + 5 = 3d + 15$

10 years

Example 8.3 Naina has twice as many cows as Gauri. If Naina gave Gauri 3 cows, they would have equal numbers. How many does each have?

SOLUTION

$2g - 3 = g + 3$ gives $g = 6$

Gauri 6 and Naina 12

Example 8.4 Each time a man walks round a tree his coins double, and he must then pay 16 coins. After three rounds he has nothing left. How many coins did he start with?

SOLUTION

$x \to 2x - 16 \to 4x - 48 \to 8x - 112$, and $8x - 112 = 0$

14 coins

Example 8.5 A food cart costs ₹5000 a day to rent and each dosa costs ₹10 to make. Dosas sell at ₹50. How many must be sold in a day for a profit of ₹2000?

SOLUTION

$50n - 10n - 5000 = 2000$ gives $40n = 7000$

175 dosas

Example 8.6 Evaluate $\frac{1}{3}, \frac{1+3}{5+7}, \frac{1+3+5}{7+9+11}$. Why are they equal?

SOLUTION

The sum of the first n odd numbers is n^2 and of the first $2n$ is $4n^2$. So the denominator is $4n^2 - n^2 = 3n^2$.

Each fraction is $\frac{n^2}{3n^2} = \frac{1}{3}$

9 Common mistakes

- Writing a two-digit number with digits a and b as ab (a product) instead of $10a + b$.
- Forgetting brackets when a whole expression is multiplied or divided: "add 4, then divide by 2" is $\frac{2x + 4}{2}$.
- Using $a + b + c$ for the top of a 3-row pyramid. It is $a + 2b + c$.
- In calendar problems, taking the number below as $a + 1$ instead of $a + 7$.
- Not checking that the answer makes sense (a date above 31, or a fraction of a cow).

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